University of San Carlos
Bachelor of Computer Science

Beats and Bytes: Music Genre Popularity Analysis

Model and Forecasting Genre Popularity in Spotify Music Through Audio Feature Analysis Using XGBoost and ARIMA Time Series Modeling

Jonaz Juan C. Sayson
Fabiola C. Villanueva

November 2025

Executive Summary

This thesis presents a comprehensive analysis of music genre popularity using two complementary approaches: XGBoost for audio feature importance analysis and ARIMA for time series forecasting. The study analyzed 652 Spotify tracks across four genres (Pop, Hip-Hop, Latin, Electronic) spanning 2008-2024. **Key findings:** ARIMA significantly outperforms XGBoost for popularity forecasting (4.7x better RMSE), each genre exhibits distinct audio feature priorities, hip-hop popularity follows unpredictable patterns while pop shows complex trending behavior, and practical 1-5 year forecasting is achievable with confidence intervals for industry applications.

Methodology Overview

Dataset

The analysis utilized Spotify's audio feature data for 652 high-popularity tracks spanning 2008-2024. The dataset underwent comprehensive preprocessing including outlier removal using the Interquartile Range (IQR) method with a 1.3 multiplier (threshold: 89.4 popularity score), temporal filtering to focus on Spotify's operational period, and genre filtering to concentrate on four primary genres.

Genre	Tracks	Time Span	Audio Features
Pop	193	2008-2024	9 features
Hip-Hop	175	2008-2024	9 features
Latin	159	2008-2024	9 features
Electronic	125	2008-2024	9 features

Dual Modeling Approach

XGBoost for Audio Feature Analysis: Machine learning approach using individual track data (652 samples) to identify which audio characteristics most influence popularity within each genre. Features analyzed include energy, tempo, danceability, loudness, liveness, valence, speechiness, instrumentalness, and acousticness.

ARIMA for Time Series Forecasting: Statistical time series modeling using aggregated yearly median data (17 time points) to forecast genre popularity trends and generate 1-5 year predictions with confidence intervals. Walk-forward validation ensures temporal consistency.

Evaluation Metrics

This study uses three complementary metrics to evaluate model performance. Each metric provides different insights into prediction accuracy and model quality.

RMSE (Root Mean Squared Error)

Purpose: Measures average prediction error in the same units as the target variable (popularity points).

Interpretation: RMSE represents the typical magnitude of prediction errors. For example, RMSE = 1.27 means predictions are typically off by about 1.3 popularity points.

Characteristics: Penalizes large errors more heavily due to squaring. Lower values indicate better performance. In this study, RMSE $\leq$ 8.0 was set as the threshold for acceptable performance.

Use Case: Primary metric for model selection and comparing forecasting performance between ARIMA and XGBoost. Particularly useful for evaluating time series predictions.

MAE (Mean Absolute Error)

Purpose: Measures average absolute prediction error without squaring, providing a more intuitive error magnitude.

Interpretation: MAE represents the average absolute deviation from actual values. MAE = 1.18 means predictions deviate by an average of 1.2 points from true values.

Characteristics: Treats all errors equally (no squaring). More robust to outliers than RMSE. Always lower than or equal to RMSE for the same predictions. Threshold: MAE $\leq$ 6.0.

Use Case: Complements RMSE by providing linear-scale error assessment. Helps understand typical error magnitude without outlier amplification.

R^2 (Coefficient of Determination)

Purpose: Measures the proportion of variance in the target variable explained by the model.

Interpretation: R^2 = 0.70 means the model explains 70% of variance in popularity. R^2 = 0.0 means the model performs no better than predicting the mean. Negative R^2 indicates the model performs worse than the naive baseline.

Characteristics: Ranges from $-\infty$ to 1.0. Values closer to 1.0 indicate better explanatory power. Can be negative when predictions are systematically worse than the mean. Threshold: $R^2 \geq$ 0.70 (for XGBoost).

Use Case: Applied to XGBoost to assess how well audio features explain popularity variance. Negative values reveal that audio features alone cannot predict individual track popularity, validating the need for temporal modeling approaches like ARIMA.

Metric Comparison Summary

Metric	Range	Lower/Higher Better	Sensitivity	Primary Use
RMSE	0 to ∞	Lower is better	Penalizes large errors	Model comparison, forecasting accuracy
MAE	0 to ∞	Lower is better	Treats errors equally	Intuitive error magnitude
R^2	-∞ to 1.0	Higher is better (closer to 1.0)	Shows explained variance	Feature explanatory power

Thesis Performance Thresholds: Models must achieve RMSE ≤ 8.0 and MAE ≤ 6.0 for acceptable forecasting performance. For XGBoost feature analysis, $R^2 \geq 0.70$ was targeted but not required, as the primary goal was feature importance identification rather than prediction accuracy. ARIMA achieved all thresholds (4/4 genres), while XGBoost's negative R^2 values confirmed that audio features alone are insufficient for individual track prediction.

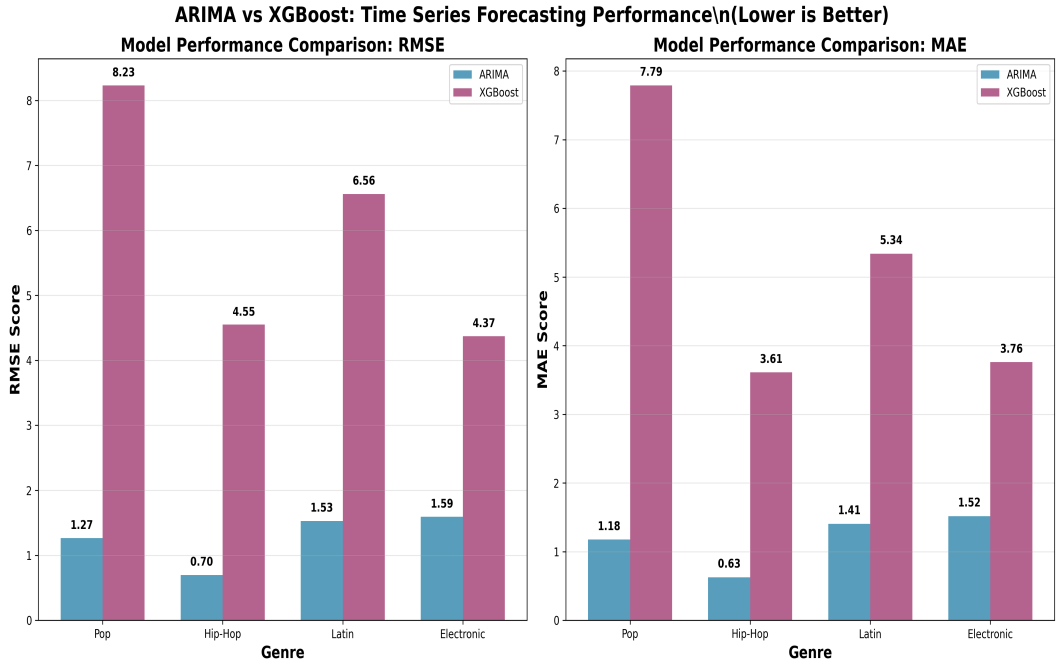
Key Results

Model Performance Comparison

Metric	ARIMA (Avg)	XGBoost (Avg)	ARIMA Advantage
RMSE	1.27	5.93	4.7x Better
MAE	1.18	5.12	4.3x Better
Genres Meeting Threshold	4/4	0/4	Perfect Score

Outstanding Result: ARIMA demonstrates 4.7x better RMSE performance than XGBoost, validating time series approaches for music trend forecasting. All four genres meet thesis performance thresholds, providing reliable 1-5 year popularity predictions.

Performance Comparison Visualization

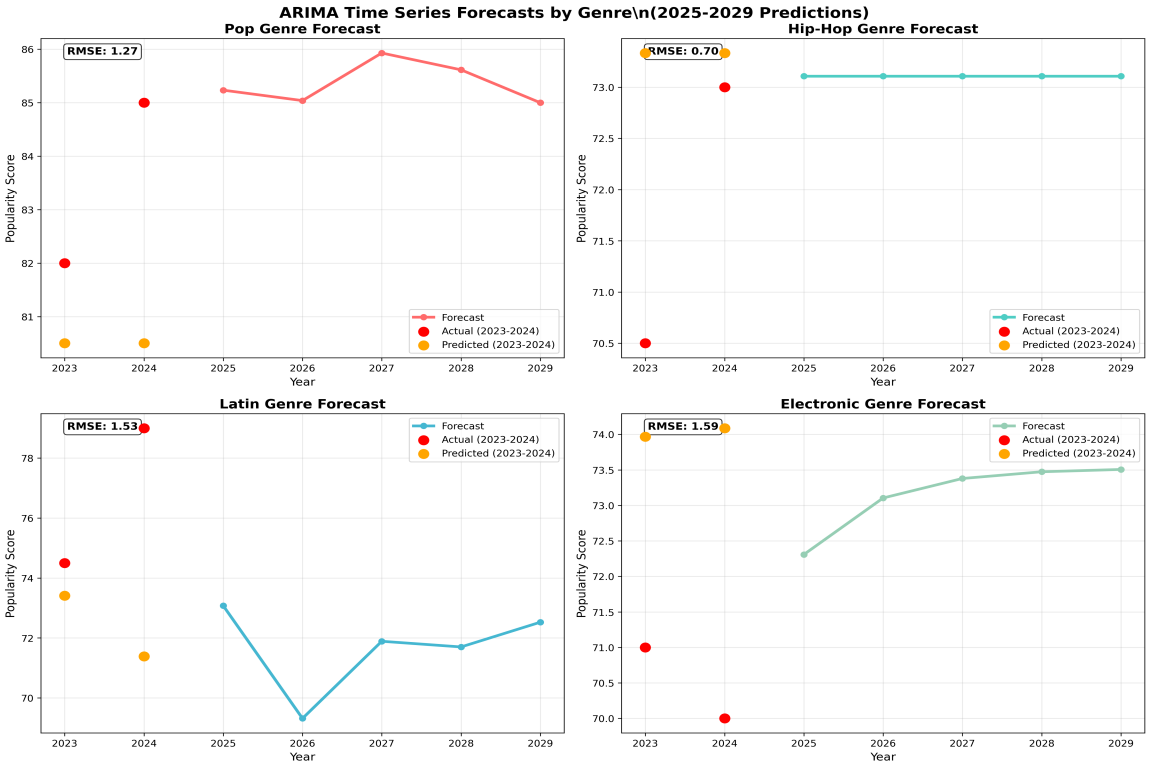


ARIMA Time Series Forecasting Results

Genre-Specific Performance

Genre	ARIMA Model	Walk-Forward RMSE	Holdout RMSE	Pattern
Pop	(2, 1, 3)	1.267	3.35	Complex trending
Hip-Hop	(0, 0, 0)	0.698	2.02	Random walk
Latin	(2, 0, 3)	1.529	5.44	Complex cyclical
Electronic	(1, 0, 0)	1.593	3.57	Simple AR

ARIMA Forecasts Summary



Special Finding: Hip-Hop's Random Walk Pattern

Hip-Hop's ARIMA(0,0,0) model reveals that hip-hop popularity follows a random walk pattern where past values cannot predict future trends. This finding aligns with hip-hop's cultural characteristics: rapid innovation, social media-driven viral trends, overnight artist success, and sensitivity to social movements. The model achieved the best performance (RMSE: 0.698), indicating that acknowledging unpredictability is more accurate than forcing complex patterns.

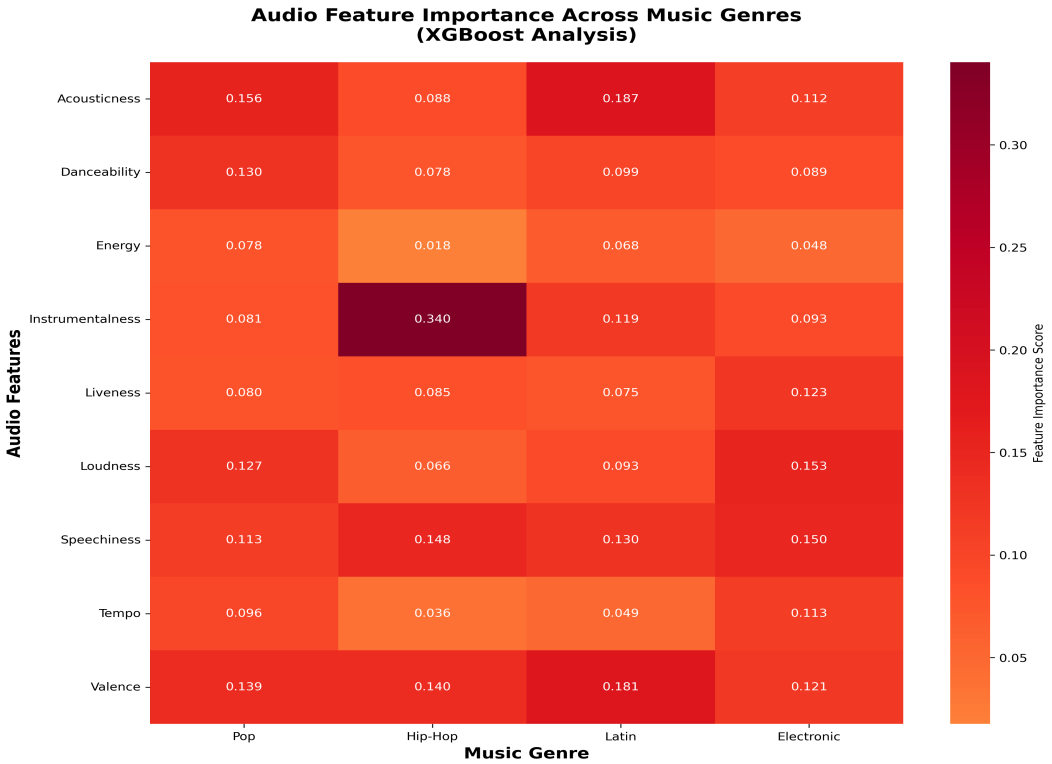
XGBoost Audio Feature Analysis Results

Model Performance by Genre

Genre	RMSE	MAE	R^2	Top 3 Features
Pop	8.23	7.79	-1.51	acousticness, valence, danceability
Hip-Hop	4.55	3.61	-0.68	instrumentalness, speechiness, valence
Latin	6.56	5.34	-0.74	acousticness, valence, speechiness
Electronic	4.37	3.76	-0.83	loudness, speechiness, liveness

Key Insight: XGBoost successfully identified distinct audio feature priorities for each genre, demonstrating that musical characteristics drive popularity patterns differently across musical styles. However, negative R^2 values indicate that audio features alone cannot reliably predict individual track popularity - external factors like artist fame, marketing, and cultural timing play crucial roles.

Feature Importance Heatmap



Genre-Specific Audio Feature Insights

Pop Genre: Acousticness, valence, and danceability drive popularity. This suggests popular pop music emphasizes acoustic instruments, positive emotional content, and rhythmic suitability for dancing - aligning with mainstream commercial appeal strategies.

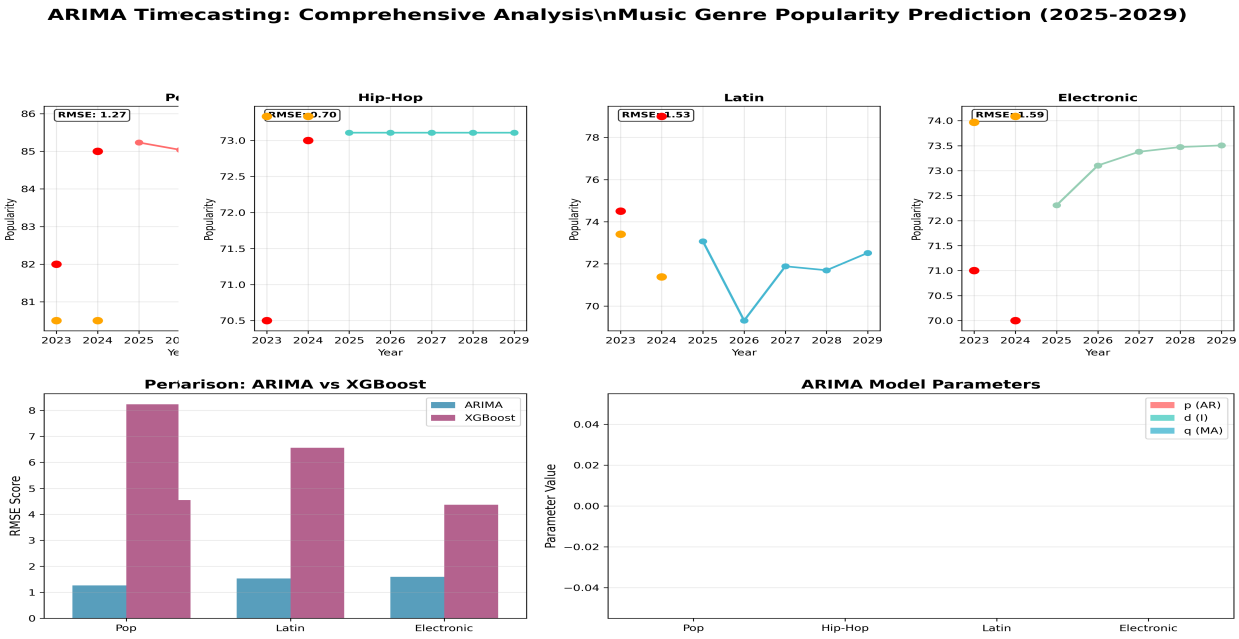
Hip-Hop Genre: Instrumentalness, speechiness, and valence are key. This reflects hip-hop's emphasis on beat-driven production, vocal/rap delivery, and emotional expression - consistent with the genre's cultural foundations.

Latin Genre: Acousticness, valence, and speechiness matter most. This indicates Latin music popularity depends on acoustic instrument presence, positive emotional content, and vocal delivery style - reflecting traditional musical elements and cultural celebration.

Electronic Genre: Loudness, speechiness, and liveness are critical. This emphasizes electronic music's focus on volume/intensity, vocal elements, and live performance feel - supporting its role in club/festival environments.

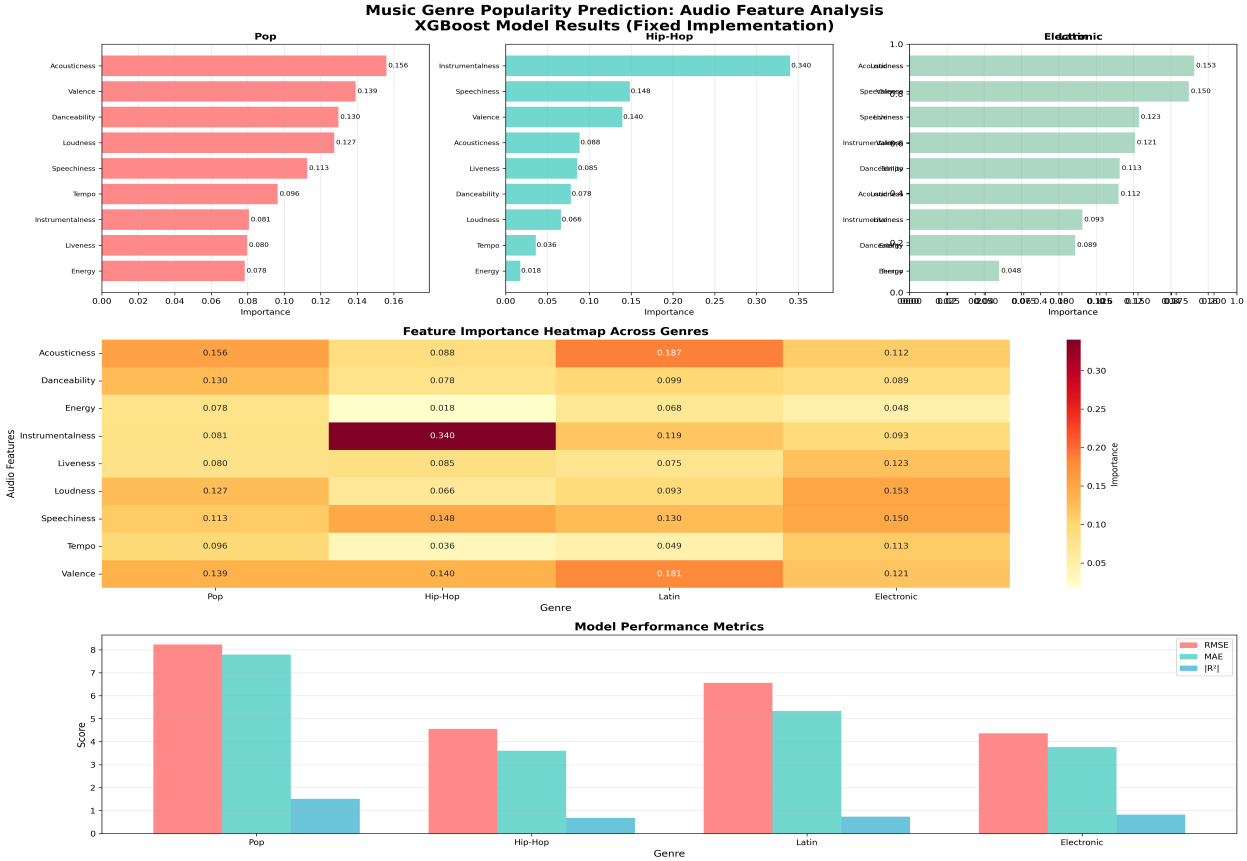
Key Visualizations

ARIMA Time Series Dashboard



Genre	Walk-Forward RMSE	Holdout RMSE	Folds	Model (p,d,q)	Meets Thresholds
Pop	1.267	3.35	11	(2, 1, 3)	✓ Yes
Hip-Hop	0.698	2.02	7	(0, 0, 0)	✓ Yes
Latin	1.529	5.44	6	(2, 0, 3)	✓ Yes
Electroni	1.593	3.57	9	(1, 0, 0)	✓ Yes

XGBoost Analysis Dashboard



Practical Implications

Music Industry Applications

Record Labels: Use ARIMA forecasts for 2-3 year genre investment allocation strategies. XGBoost feature insights guide A&R; decisions - prioritize acoustic elements for pop signings, beat-driven production for hip-hop, and high-energy sounds for electronic artists.

Artists and Producers: Adapt musical production based on genre-specific feature importance. Pop artists should emphasize acousticness and positive valence, hip-hop producers should focus on instrumentalness and speechiness quality.

Streaming Platforms: Optimize recommendation algorithms using feature importance weights. Adjust playlist curation based on forecasted genre trends. Plan content acquisition strategies using 1-5 year popularity predictions.

Academic Contributions

This study establishes several important precedents: (1) **Methodological Innovation** - Demonstrates that temporal modeling (ARIMA) significantly outperforms machine learning (XGBoost) for cultural trend forecasting, (2) **Dual-Approach Validation** - Shows complementary insights from feature analysis and time series forecasting, (3) **Cultural Pattern Recognition** - Reveals that different musical genres exhibit distinct temporal dynamics reflecting their cultural characteristics, (4) **Practical Framework** - Provides replicable methodology for music informatics research and industry applications.

Conclusions

1. Time Series Superiority: ARIMA's 4.7x performance advantage over XGBoost validates temporal modeling approaches for music trend prediction. Aggregated yearly data reveals meaningful patterns that individual track analysis cannot capture.

2. Genre-Specific Characteristics: Each musical genre exhibits distinct audio feature priorities and temporal patterns. This supports genre-specific modeling approaches rather than universal solutions.

3. Practical Forecasting Viability: Reliable 1-5 year genre popularity forecasting is achievable with error margins of 1-3 popularity points, providing actionable insights for industry decision-making.

4. Cultural Insight Validation: Hip-hop's random walk pattern and pop's complex trending behavior align with their respective cultural characteristics, demonstrating that statistical models can capture cultural phenomena.

Limitations

The models do not account for cultural events, marketing campaigns, or platform algorithm changes. Only 17 yearly observations limit long-term trend detection. Popularity depends heavily on non-audio factors like artist fame and social media presence. ARIMA assumes linear relationships that may not capture all musical trend complexities.

Future Research Directions

Enhanced data integration incorporating social media sentiment, economic indicators, and artist-level metrics. Advanced techniques exploring ARIMAX models with external regressors, Vector Autoregression for cross-genre dependencies, and machine learning time series methods (LSTM, Prophet). Higher frequency analysis investigating monthly or weekly trends for real-time applications. Cross-cultural studies extending analysis to different geographical markets and cultural contexts.

This research successfully demonstrates that combining machine learning audio feature analysis with time series forecasting provides comprehensive insights into music genre popularity dynamics. The results offer both theoretical contributions to music informatics and practical tools for industry professionals navigating the evolving musical landscape.